

Dyslexia Research

1. What are the most common reading and writing challenges faced by people with dyslexia, and how can they be effectively simulated in a digital environment?

- 1.1. Common Challenges:

Individuals with dyslexia often struggle with reading comprehension due to difficulties in decoding words, leading to challenges in accurately and fluently reading text. This can hinder their ability to understand complex passages and grasp main ideas. devdis.com.

- 1.2. Digital Simulations:

Some digital simulations attempt to replicate the reading challenges faced by individuals with dyslexia. For example, a dyslexia simulator demonstrates how text can appear to move around the screen, making reading difficult. [AbilityNet](#)

However, it's important to note that not all simulations accurately represent the dyslexic experience. Some argue that certain online simulations, while well-intentioned, may not truly capture what it's like to have dyslexia. [International Dyslexia Association](#)

2. How do different types of dyslexia (e.g., phonological, surface, visual) affect reading and comprehension, and how can these be represented in a gamified format?

- 2.1. Types of Dyslexia and Their Effects:

- **Phonological Dyslexia:** This type involves difficulties in connecting sounds to their written forms, leading to challenges in decoding unfamiliar words and slow reading. [arXiv](#)
- **Surface Dyslexia:** Individuals may struggle with recognizing whole words by sight, relying heavily on phonetic decoding, which can affect reading fluency.
- **Visual Dyslexia:** This form can involve difficulties with visual processing, such as letter and word reversals, impacting reading accuracy. [Forbrain](#)

- 2.2. Gamified Representations:

Gamification has been explored as a supportive tool for children with dyslexia. Research indicates that video games can improve letter-to-speech sound mapping (phonological decoding) and, consequently, reading abilities. [MDPI](#)

A virtual reality (VR) serious game has been developed to increase empathy towards students with phonological dyslexia. In this game, players must create a potion by following a recipe written in an alphabet designed to replicate the reading difficulties experienced by individuals with dyslexia. [arXiv](#)

3. What other problems do people with dyslexia face when accessing the internet and social media?
 - 3.1. **Reading and Comprehension Difficulties:** The text-heavy nature of many websites and social media platforms can be overwhelming for individuals with dyslexia, leading to difficulties in reading and comprehending content.
 - 3.2. **Writing and Expression Challenges:** Crafting posts, comments, or messages can be daunting due to spelling difficulties and concerns about negative feedback, potentially hindering self-expression and participation in online discussions. [Engineering at Meta](#)
 - 3.3. **Social and Emotional Impacts:** The frustration from reading and writing challenges can lead to feelings of anxiety and reduced self-esteem, affecting social interactions both online and offline.
 - 3.4. **Risk of Digital Exclusion:** Inaccessible digital environments and lack of assistive technologies can increase the risk of digital exclusion for individuals with dyslexia, limiting their participation in online activities. [British Dyslexia Association](#)
 - 3.5. **Bullying and Social Isolation:** Individuals with dyslexia may be at a higher risk of bullying, both in school and online, particularly when their reading and writing difficulties are more visible. [Ineqe Safeguarding Group](#)
 - 3.6. **Navigational Challenges:** Complex website layouts and navigation structures can be confusing, making it difficult for individuals with dyslexia to find information or use certain features effectively. [British Dyslexia Association](#)
 - 3.7. **Password Management Difficulties:** Remembering and managing passwords can be particularly challenging, posing security risks and access issues.

Addressing these challenges requires implementing universal design principles, offering assistive technologies, and fostering supportive online communities to enhance accessibility and inclusivity for individuals with dyslexia. [British Dyslexia Association](#)

4. Are there other things like software or physical tools that helps people with dyslexia browse the internet and social media?

4.1. Software Tools:

- **Text-to-Speech (TTS) Software:** These applications read digital text aloud, aiding comprehension and reducing the strain of reading. They are particularly useful for navigating text-heavy websites and social media platforms.
- **Speech-to-Text (Dictation) Software:** By converting spoken words into written text, these tools assist individuals with dyslexia in composing messages, posts, or emails, thereby minimizing spelling and grammar concerns. [AbilityNet](#)
- **Dyslexia-Friendly Fonts and Display Adjustments:** Certain browser extensions and applications allow users to customize font styles, sizes, and background colors to enhance readability. Implementing dyslexia-friendly fonts and adjusting text settings can significantly improve the online reading experience. [Accessibly](#)
- **Assistive Reading Applications:** Apps like Augmenta11y utilize augmented reality to present printed text in a dyslexia-friendly format on a user's device, allowing for personalized font and color settings to improve readability. [Wikipedia](#)

4.2. Physical Tools:

- **Smart Pens:** These devices digitize handwritten notes and often include audio recording features, enabling users to review and organize information more effectively. [IDA Oregon](#)
- **Reading Guides and Overlays:** Physical tools such as colored overlays or reading strips can help individuals focus on specific lines of text, reducing visual stress and improving reading accuracy.

4.3. Integrated Accessibility Features:

- Modern operating systems and browsers offer built-in accessibility options, including voice recognition, magnification, and high-contrast modes, which can be tailored to support users with dyslexia. [AbilityNet](#). By leveraging these technologies and tools, individuals with dyslexia can navigate the internet and social media platforms more effectively, fostering a more inclusive digital environment.

5. How These Findings Influenced the Project

This research directly shaped the development of an **interactive website designed to simulate dyslexia** for users. The goal was to **move beyond theoretical explanations** and provide a **hands-on experience** illustrating how dyslexia affects reading and navigation in digital spaces.

Based on research findings:

- **Text Distortion Simulation:** Implementing fluctuating letters, inconsistent spacing, and movement effects to mimic visual dyslexia symptoms.
- **Reading Interference Mechanics:** Users experience shifting words and distractions to simulate real cognitive difficulties in processing text.
- **Navigation Barriers:** The website includes usability challenges to show how poorly designed layouts affect accessibility for dyslexic users.
- **Customizable Accessibility Options:** Users can adjust text formats, fonts, and contrast settings to demonstrate the impact of inclusive digital design.

These elements transformed the project into an **empathy-building tool** that offers a more authentic representation of dyslexic challenges in digital environments.

6. Team Organization & Workflow

The project was structured around **collaborative research and development**:

- **Research & Data Collection:** Investigating dyslexia symptoms, accessibility concerns, and existing digital barriers.
- **Design & Implementation:** Translating research insights into an interactive web simulation.
- **Testing & Feedback:** Gathering responses from individuals with dyslexia to improve accuracy and effectiveness.

Throughout development, **team communication and organization** played a key role in ensuring each element of the website accurately reflected real-world experiences. Adjustments were made based on **expert consultations** to ensure authenticity in the digital representation of dyslexia.

7. Conclusion

Dyslexia presents a wide range of challenges in reading, writing, and digital navigation. This research provided the foundation for an **interactive website simulation** designed to help users experience the difficulties firsthand, fostering **empathy and awareness**.

By refining digital accessibility, implementing assistive tools, and promoting inclusive design, this project contributes to **a more supportive digital landscape** for individuals with dyslexia.